

Cross-Distribution Deployment

Fedora, RHEL, AlmaLinux, Ubuntu — One Script, Any Host

Auto-Detect | [deploy-remote.sh](#) | [hivex Rebuild](#) | [SPICE/VNC](#) | [Proprietary \(HyperSDK\)](#)

github.com/ssahani/hyper2kvm

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1. The Challenge

Every Linux distribution ships different packages, paths, and Python versions. hyper2kvm handles all of them.

Component	Fedora 43	RHEL 8.8	AlmaLinux 9	Ubuntu 24.04
Python	3.13	3.6	3.9	3.12
QEMU path	/usr/bin/qemu-system-x86_64	/usr/libexec/qemu-kvm	/usr/libexec/qemu-kvm	/usr/bin/qemu-system-x86_64
Graphics	SPICE + QXL	VNC only	VNC only	SPICE + QXL
hivex Python	dnf package	dnf package	source rebuild	source rebuild
libguestfs	dnf package	dnf package	dnf package	apt package
virtio-win	dnf package	subscription	manual download	manual download
OVMF path	/usr/share/edk2/ovmf/	/usr/share/OVMF/	/usr/share/OVMF/	/usr/share/OVMF/
supermin	dnf package	dnf package	dnf package	apt package

Pain Points

- Python 3.12 drops `distutils` — breaks older `setup.py` builds
- RHEL/AlmaLinux ship no SPICE libraries — must use VNC
- QXL video model fails without SPICE — must use `virtio-gpu`
- `hivex` Python bindings missing on newer distros — source build required
- `virtio-win` ISO not packaged on non-Red Hat systems
- `tmpfs /run` directory missing `h2kvmctl` runtime dirs

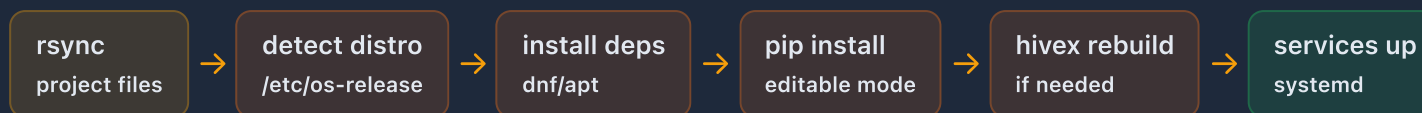
Solution: Auto-Everything

- Auto-detect QEMU binary from 3 known paths
- Auto-detect SPICE availability via `libspice-server.so`
- Auto-select QXL (SPICE) vs `virtio-gpu` (VNC)
- Auto-rebuild `hivex` from source when needed
- Auto-download `virtio-win` ISO if missing
- Auto-create runtime directories via `tmpfiles.d`

Goal: Run `deploy-remote.sh` on any supported distro and get a fully working migration environment. No manual package hunting, no path editing, no configuration files.

2. deploy-remote.sh

One-command deployment: rsync the project, install dependencies, build hivex, configure services.



What It Does

- Detects distro family: RHEL, Fedora, Debian/Ubuntu
- Installs system packages (qemu-kvm, libvirt, libguestfs, etc.)
- Installs Python packages via pip (editable mode)
- Rebuilds hivex from source if Python bindings missing
- Downloads virtio-win ISO to `/usr/share/virtio-win/`
- Installs h2kvmctl to `/usr/local/sbin/`
- Installs h2kweb systemd service
- Creates tmpfiles.d entry for `/run/hyper2kvm`

Usage

```
# Deploy to remote server
./scripts/deploy-remote.sh \
  --host 185.165.240.5 \
  --user root \
  --key ~/.ssh/id_ed25519

# Deploy locally
./scripts/deploy-remote.sh --local

# With custom options
./scripts/deploy-remote.sh \
  --host 10.0.1.50 \
  --port 2222 \
  --no-web          # skip h2kweb
  --no-virtio-win  # skip ISO download
```

Distro-Specific Package Lists

Fedora / RHEL / AlmaLinux

- qemu-kvm, libvirt, virt-install
- libguestfs, libguestfs-tools
- python3-hivex, hivex-devel
- python3-libguestfs
- python3-augeas
- nbdkit, qemu-img
- supermin

Ubuntu / Debian

- qemu-system-x86, libvirt-daemon-system
- libguestfs-tools, libguestfs-dev
- libhivex-dev, python3-hivex
- python3-guestfs
- python3-augeas
- nbdkit, qemu-utils
- supermin

Common (pip)

- pyyaml, jinja2
- rich, click
- requests
- libvirt-python
- hyper2kvm (editable)
- h2kweb frontend build

3. Auto-Detection

Runtime detection of host capabilities ensures correct configuration without manual tuning.

SPICE / QXL Availability

- Check: `ldconfig -p | grep libspice-server`
- SPICE present: use QXL video + SPICE graphics
- SPICE absent: use virtio-gpu video + VNC graphics
- Fedora/Ubuntu: typically have SPICE
- RHEL/AlmaLinux: typically VNC-only
- Domain XML generated with correct model automatically

QEMU Binary Paths

- Priority 1: `/usr/bin/qemu-system-x86_64` (Fedora, Ubuntu)
- Priority 2: `/usr/libexec/qemu-kvm` (RHEL, AlmaLinux)
- Priority 3: `/usr/bin/qemu-kvm` (older systems)
- Verified: file exists and is executable
- Used in domain XML `<emulator>` element
- Error if no QEMU binary found on system

libguestfs Appliance

- Scan: `/usr/lib64/guestfs/`
- Scan: `/usr/lib/guestfs/`
- Set LIBGUESTFS_PATH if needed
- supermin rebuild if stale
- Kernel + initrd verification

OVMF Firmware

- `/usr/share/edk2/ovmf/` (Fedora)
- `/usr/share/OVMF/` (RHEL/Ubuntu)
- Prefer secboot variant if present
- NVRAM template path auto-set
- BIOS fallback for MBR guests

VirtIO-Win ISO

- `/usr/share/virtio-win/` (packaged)
- `/var/lib/hyper2kvm/` (downloaded)
- Auto-download from [fedorapeople.org](https://www.fedorapeople.org)
- Version detection from filename
- Attached as CD-ROM in domain XML

All detection happens at runtime. The same h2kvmctl binary and Python code runs identically on Fedora 43 and AlmaLinux 9 — auto-detection adapts the output (domain XML, graphics, firmware) to what the host actually provides.

SPICE auto-detect

QEMU path scan

libguestfs appliance

OVMF firmware

VirtIO ISO auto-download

4. Dependency Resolution

Handling the hard cases: hivex source rebuild, libguestfs symlinks, NBD module loading.

hivex Source Rebuild

- Required when: Python 3.12+ and no distro python3-hivex package
- AlmaLinux 9 (Python 3.9): package available, no rebuild
- Ubuntu 24.04 (Python 3.12): must rebuild from source
- Build deps: hivex-devel, python3-devel, gcc, automake
- Script: `./configure --prefix=/usr && make && make install`
- Installs to system Python site-packages

libguestfs Symlink

- Some distros install to `/usr/lib64/` but Python checks `/usr/lib/`
- `deploy-remote.sh` creates symlink if needed
- `LIBGUESTFS_PATH` environment variable set in service file
- supermin appliance rebuilt after package updates
- Appliance cache at `/var/tmp/.guestfs-*`

supermin

- Required for libguestfs appliance
- Builds minimal Linux kernel + initrd
- Auto-rebuild on kernel updates
- Cache: `/var/tmp/.guestfs-0/`
- `install-deps.sh` includes supermin

python-augeas

- Used for Linux guest config editing
- Fedora/RHEL: `python3-augeas`
- Ubuntu: `python3-augeas`
- Edits `/etc/fstab`, `grub.cfg`, network
- Lens-based config file parsing

NBD Module

- `modprobe nbd max_part=8`
- Required for qemu-nbd disk mount
- Persistent via `/etc/modules-load.d/nbd.conf`
- Options via `/etc/modprobe.d/nbd.conf`
- Verified during readiness checks

Build Matrix

Distro	hivex	libguestfs	augeas	NBD	Status
Fedora 43	✓ Package	✓ Package	✓ Package	✓ Module	✓ Ready
RHEL 8.8	✓ Package	✓ Package	✓ Package	✓ Module	✓ Ready
AlmaLinux 9	Source	✓ Package	✓ Package	✓ Module	✓ Ready
Ubuntu 24.04	Source	✓ Package	✓ Package	✓ Module	✓ Ready
Ubuntu 22.04	Source	✓ Package	✓ Package	✓ Module	✓ Ready

5. Runtime Infrastructure

Directories, locking, tmpfiles.d, daemon configuration, and service management.

/run/hyper2kvm

- Created by tmpfiles.d on boot: `d /run/hyper2kvm 0755 root root -`
- Also created by h2kweb service on startup
- Stores runtime PID files and lock files
- NBD device lock files prevent concurrent access
- Conversion progress files for WebSocket streaming
- Cleaned on reboot (tmpfs)

NBD Locking

- Lock file per NBD device: `/run/hyper2kvm/nbd0.lock`
- Prevents two conversions from using same `/dev/nbd*`
- Acquired before `qemu-nbd --connect`
- Released after `qemu-nbd --disconnect`
- Stale lock detection and cleanup on startup
- Supports up to 16 concurrent NBD devices

Conversion Directories

- Input: `/var/lib/hyper2kvm/incoming/`
- Output: `/var/lib/hyper2kvm/converted/`
- Temp: `/tmp/hyper2kvm-*` (16GB tmpfs)
- Configurable via CLI or daemon config
- Auto-cleanup of temp files on completion

Daemon Config

```
# /etc/hyper2kvm/daemon.yaml
watch_dir: /var/lib/hyper2kvm/incoming
output_dir: /var/lib/hyper2kvm/converted
auto_deploy: true
concurrency: 2
log_level: info
```

systemd Services

- `h2kweb.service` — web dashboard
- `h2kvmctl-daemon.service` — watch dir
- Both: `Restart=on-failure`
- Both: `After=libvirt.service`
- Journal integration for logging

File System Layout

Path	Purpose
<code>/usr/local/sbin/h2kvmctl</code>	Main CLI binary (editable install symlink)
<code>/usr/bin/zkvm</code>	TUI binary (Go, Bubble Tea)
<code>/usr/bin/h2kweb</code>	Web dashboard binary (Go + embedded frontend)
<code>/run/hyper2kvm/</code>	Runtime: PID, locks, progress (tmpfs)
<code>/var/lib/hyper2kvm/</code>	Persistent: converted images, virtio-win ISO
<code>/etc/hyper2kvm/</code>	Configuration: daemon.yaml

6. Migration Readiness Dashboard

19 automated checks verify that the host is fully ready to migrate VMs. All green = ready to migrate.

#	Check	What It Verifies	Fedora 43	AlmaLinux 9
1	KVM acceleration	<code>/dev/kvm</code> present and accessible	✓	✓
2	QEMU binary	qemu-system-x86_64 or qemu-kvm found	✓	✓
3	libvirt daemon	libvirtd running and socket accessible	✓	✓
4	qemu-img	Image conversion tool available	✓	✓
5	qemu-nbd	NBD server for disk mounting	✓	✓
6	NBD kernel module	<code>lsmod grep nbd</code> loaded	✓	✓
7	libguestfs	Python bindings importable	✓	✓
8	hivex	Registry editing bindings available	✓	✓ (rebuilt)
9	augeas	Config file editing available	✓	✓
10	OVMF firmware	UEFI firmware files found	✓	✓
11	virtio-win ISO	Windows VirtIO drivers present	✓	✓
12	virsh	Domain management CLI available	✓	✓
13	virt-install	VM creation tool available	✓	✓
14	Network bridge/NAT	Default libvirt network active	✓	✓
15	Disk space	Sufficient free space for conversion	✓	✓
16	Runtime directory	<code>/run/hyper2kvm</code> exists and writable	✓	✓
17	supermin appliance	libguestfs appliance built and cached	✓	✓
18	ntfs-3g	NTFS mount support for Windows guests	✓	✓
19	govc (optional)	VMware vSphere CLI for remote migration	✓	✓

One Script, Any Host, All Green

deploy-remote.sh handles every distro difference automatically. After deployment, all 19 checks pass.

Fedora, RHEL, AlmaLinux, Ubuntu — same binary, same config, same results.

19/19 checks passing

4 distros supported

Auto hivex rebuild

SPICE/VNC auto-select

One-command deploy

doctor.sh diagnostics

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